WEB APPLICATION DEVELOPMENT — CS2682

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1. Mariam Sana Tariq — AF25LHB4358

Project: TalentPulse — AI-Style Recruitment Intelligence SaaS
An end-to-end hiring platform for HR departments. Companies post jobs, the system auto-scores incoming applicants based on skill-keyword matching against their resumes, tracks candidate pipelines on a Kanban-style interview board, and gives hiring managers a live dashboard showing pipeline bottlenecks, offer acceptance rates, and time-to-hire per department.

- **Industry Role:** Replaces manual HR screening — directly relevant to every company doing hiring in 2026
- **Free APIs:** GitHub REST API (candidate technical profile linking), Hunter.io free tier (email verification of applicants), Chart.js (funnel analytics), EmailJS free tier (automated interview invite emails), FullCalendar.js (interview scheduling calendar)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase Realtime DB
- **SaaS Architecture:** Company-level tenant accounts, HR Manager / Recruiter / Interviewer roles, job-wise pipelines
- **Real-Time Element:** Live applicant counter updates the moment a new application is submitted via Firebase listener, Kanban card moves trigger AJAX status updates visible to all logged-in recruiters simultaneously, interview calendar conflicts detected and flagged instantly

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2. Eman Fatima — AF25LHB4359

Project: BrandSentinel — Brand Reputation & Crisis Intelligence SaaS
Marketing and PR teams monitor their brand name across Reddit, HackerNews, and live news sources in real-time. The platform scores each mention as positive, neutral, or negative using keyword-weight scoring, tracks sentiment trends over 30 days, and fires a crisis alert dashboard when a negative spike exceeds a threshold — giving the team time to respond before damage escalates.

- **Industry Role:** Every brand, startup, and public company needs reputation monitoring — a multi-billion dollar market
- **Free APIs:** Reddit API (real-time public post/comment search — free), HackerNews Algolia API (completely free — real-time search), NewsAPI.org free developer tier (news mentions), Chart.js (sentiment trend graphs), Supabase free tier (real-time alert delivery)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Brand workspaces, team member roles, keyword watchlists, alert escalation levels

- **Real-Time Element:** Incoming mentions stream live into the dashboard via AJAX polling every 90 seconds, sentiment score meter animates on new data, crisis alert banner appears with sound notification when threshold crossed

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3. M. Saib Bashin — AF25LHB4537

Project: LogiTrack — Supply Chain Shipment Visibility SaaS
Businesses with multi-vendor supply chains track every shipment from supplier dispatch to warehouse arrival on a live map. Each shipment has status milestones (dispatched, in-transit, at-customs, delivered), delay risk scoring based on overdue timelines, and supplier performance scorecards. Procurement managers get a disruption alert feed when any active shipment goes overdue.

- **Industry Role:** Supply chain visibility is the top priority for manufacturing, retail, and logistics companies post-pandemic disruptions
- **Free APIs:** Leaflet.js + OpenStreetMap (free live map, no API key), OSRM API (free routing/distance calculation), Nominatim (free geocoding — convert city names to coordinates), Chart.js (supplier performance charts), EmailJS free tier (delay alert emails)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Company tenants, Supplier / Warehouse / Manager roles, multi-shipment tracking per company
- **Real-Time Element:** Live map pin moves as shipment status updates submitted by suppliers, overdue shipments flash red on dashboard, supplier delay rate recalculates live as new data arrives

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4. Muhammad Bilal — AF25LHB4549

Project: PriceIntel — E-Commerce Competitor Price Intelligence SaaS
Retailers and online sellers track competitor product prices fetched via public data, receive instant alerts when a competitor drops price below their threshold, run margin calculators, and get a market positioning score showing whether they are priced above, at, or below market average across their catalog categories.

- **Industry Role:** Dynamic pricing intelligence is used by every major retailer — Amazon, Daraz, and all e-commerce platforms use it internally. SME sellers need affordable access to the same intelligence
- **Free APIs:** Open Exchange Rates free tier (multi-currency margin calculations), Chart.js (price history trend lines), Alpha Vantage free tier (commodity raw material pricing reference for cost calculations), jsPDF (competitor analysis PDF export)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Retailer accounts, product catalog management, competitor product mappings, alert rule engine per SKU

- **Real-Time Element:** Price comparison dashboard refreshes automatically when any tracked product data updates via AJAX, alert badge counter increments live, margin impact calculator updates numbers as user adjusts pricing

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5. Ali Azam — AF25LHB4550

Project: CarbonLedger — Corporate ESG & Carbon Footprint Compliance SaaS
Companies facing ESG reporting requirements log their energy consumption, business travel, fleet fuel usage, and supply chain emissions. The platform calculates their carbon footprint by emission category, tracks progress toward reduction targets, benchmarks them against industry averages, and generates board-ready sustainability reports. Carbon offset recommendations with verified project links are also provided.

- **Industry Role:** ESG compliance is now legally mandatory for listed companies globally, and large corporations require ESG data from their entire supply chain — making this a tool every SME supplier will need
- **Free APIs:** World Bank Climate Change API (completely free — industry benchmark emission data), CO2 Signal API (free tier — real-time carbon intensity of electricity grids by country), Chart.js (emission breakdown charts), jsPDF (ESG report generation), Open Exchange Rates (multi-currency cost conversion)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Company accounts, department-level emission logging, yearly target setting, multi-site support
- **Real-Time Element:** Total carbon footprint counter updates live as new emission entries are submitted by department managers, progress-to-target gauge chart animates on each data entry, live carbon intensity ticker shows current grid emissions for the company's region

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6. Mubeen Ali Zulfiqar — AF25LHB4552

Project: CyberPulse — Real-Time Cybersecurity Threat Intelligence SaaS
IT security teams monitor live threat feeds, receive CVE vulnerability alerts specifically relevant to the technologies in their stack (which they register on setup), track active global cyber attacks on a threat map, log internal security incidents, and score their organization's risk posture across network, application, and human (phishing awareness) layers.

- **Industry Role:** Cybersecurity is the fastest-growing IT sector globally. Every organization needs threat intelligence but cannot afford enterprise tools like CrowdStrike. This fills the SME gap
- **Free APIs:** CIRCL CVE Search API (completely free — CVE vulnerability database), HaveIBeenPwned API (free — breach detection for company domains), AbuseIPDB API

(free tier — IP reputation checking), Shodan free tier (internet exposure scanning), Chart.js (risk trend visualization), Leaflet.js (global threat map)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Organization accounts, tech stack registration, security team roles, incident management module
- **Real-Time Element:** Live CVE feed for registered technologies auto-refreshes every 10 minutes via AJAX, new high-severity vulnerability triggers red alert banner, incident log updates in real-time for all team members via Firebase

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7. Abdul Subhan — AF25LHB4553

Project: FleetOps — Smart Fleet & Last-Mile Delivery Management SaaS
Logistics companies manage their driver fleet, assign deliveries, track routes on a live map, calculate fuel consumption per route, alert dispatchers when drivers go off-route or deliveries are overdue, and give end-customers a live delivery tracking link with ETA.

- **Industry Role:** Last-mile delivery is the backbone of e-commerce. Every courier company, food delivery service, and logistics provider needs an affordable fleet management system
- **Free APIs:** Leaflet.js + OpenStreetMap (free live map), OSRM routing API (completely free — real-time route calculation and ETA), Nominatim (free geocoding), EmailJS (customer delivery notification emails), Chart.js (driver performance analytics)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase Realtime DB (driver location simulation)
- **SaaS Architecture:** Dispatcher / Driver / Customer roles, delivery order management, route assignment, public tracking link generation per delivery
- **Real-Time Element:** Driver location pins move on the dispatcher map every 15 seconds via Firebase, customer tracking page shows live ETA countdown, overdue delivery alerts fire automatically when ETA is exceeded

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8. Ahmad Aamir — AF25LHB4554

Project: FinFlow — Business Cash Flow Intelligence & Invoice SaaS
SMEs manage their entire accounts receivable cycle: raise invoices, track payment due dates, predict upcoming cash flow gaps using trend modeling, automate payment reminder emails to clients, and get a 30-day rolling cash position forecast. Finance managers see a live treasury dashboard showing cash-in vs cash-out velocity.

- **Industry Role:** Cash flow mismanagement is the number one reason SMEs fail. Every small business owner needs this — a market of 50+ million SMEs in Pakistan alone, hundreds of millions globally

- **Free APIs:** ExchangeRate-API free tier (multi-currency invoice support), Open Exchange Rates (currency conversion), Chart.js (cash flow forecast waterfall chart), jsPDF (professional invoice PDF generation), EmailJS free tier (automated payment reminders), FullCalendar.js (payment due date calendar)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Business accounts, client directory, invoice lifecycle management, cash flow projection engine
- **Real-Time Element:** Treasury dashboard cash position total updates live as payments are marked received, overdue invoice counter increments as due dates pass, 30-day forecast chart recalculates and re-renders instantly on each transaction entry

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9. Khanzada Sarim — AF25LHB4555

Project: PropIntel — Real Estate Investment Intelligence SaaS
Real estate investors, agents, and developers track property listings across markets, run ROI and rental yield calculators for specific properties, monitor price appreciation trends by area, compare mortgage options from multiple banks, score properties on an investment attractiveness index, and generate client-facing investment proposal reports.

- **Industry Role:** Real estate is the largest asset class in Pakistan and globally. No affordable analytical SaaS exists for local property markets — massive white space opportunity
- **Free APIs:** Leaflet.js + OpenStreetMap (free property location mapping), Nominatim (free geocoding), Chart.js (price trend and ROI graphs), Open Exchange Rates (PKR/USD dual pricing), jsPDF (investment proposal report export), World Bank API (free — GDP and economic indicators for market context)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Investor / Agent / Developer role types, property portfolio management, market comparison tools, client presentation module
- **Real-Time Element:** Investment score recalculates live as user adjusts price, rent, and mortgage inputs, comparable property prices update from database on area selection, mortgage rate comparison table refreshes when bank rate inputs are edited

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10. Zarmeen Nadeem — AF25LHB4556

Project: HealthOps — Hospital Resource & Patient Flow Management SaaS
Hospitals manage bed occupancy in real-time across all wards, track OPD patient queues with estimated wait times, monitor medicine and equipment inventory with low-stock alerts, manage doctor shift schedules, and give hospital administrators a live operational dashboard showing ward utilization, patient throughput, and resource bottlenecks.

- **Industry Role:** Healthcare resource management directly affects patient outcomes. Pakistan has 1,200+ hospitals and thousands of clinics — none have affordable real-time operational dashboards
- **Free APIs:** FullCalendar.js (doctor shift scheduling), Chart.js (bed occupancy and patient flow analytics), EmailJS free tier (critical stock alert notifications), jsPDF (daily operational summary report), QRCode.js (free — patient admission QR wristband generation)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase Realtime DB
- **SaaS Architecture:** Hospital admin / Ward manager / Doctor / Pharmacist roles, ward-level bed management, OPD queue system
- **Real-Time Element:** Bed availability counter on the central dashboard updates the moment a patient is admitted or discharged via Firebase, OPD queue number display refreshes live for waiting patients, medicine stock level drops instantly when dispensary logs usage

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11. Abeeha — AF25LHB4559

Project: InfluenceIQ — Influencer Marketing Campaign ROI SaaS
Brand marketing teams discover influencers by niche and audience size using social platform APIs, launch campaigns with deliverable tracking, calculate real ROI per influencer (cost vs engagement generated), benchmark campaign performance against industry averages, and manage influencer contracts and payment milestones all in one platform.

- **Industry Role:** Influencer marketing is a $21 billion global industry growing 30% annually. Brands waste massive budgets on wrong influencers. This directly solves that problem
- **Free APIs:** Reddit API (niche community engagement data for micro-influencer discovery — free), YouTube Data API v3 (free — channel stats, subscriber count, video engagement metrics), Last.fm API (music influencer niche data), Chart.js (campaign ROI timeline), Cloudinary free tier (influencer media asset storage)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Brand / Influencer / Agency role types, campaign workspace, deliverable checklist, payment milestone tracking
- **Real-Time Element:** Campaign engagement score updates live as influencer post metrics are entered, ROI gauge chart re-renders on each data update, live budget burn tracker shows remaining campaign spend vs performance index

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12. Amna Shoukat — AF25LHB4560

Project: GovTender — Government Procurement & Public Tender Intelligence SaaS
Businesses that supply goods or services to government agencies monitor newly published tenders in real-time, receive category-based alerts for relevant opportunities, track bid submission deadlines on a compliance calendar, store their submitted bid documents, and analyze their win/loss history with competitor bid value benchmarking.

- **Industry Role:** Government procurement in Pakistan alone is worth trillions of PKR annually. Thousands of businesses miss tenders due to lack of monitoring infrastructure
- **Free APIs:** PPRA Pakistan public tender RSS/data feeds (publicly accessible), World Bank Procurement API (free — international tenders), FullCalendar.js (deadline calendar), Chart.js (win rate analytics and bid value trends), jsPDF (bid documentation package export), EmailJS (new tender alert emails)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Business accounts, category preference profiles, bid management workspace, document vault per tender
- **Real-Time Element:** New tender alert badge increments live when fresh tenders matching saved categories are detected via AJAX polling, deadline countdown timers tick in real-time on the dashboard, submitted bid tracker updates status when outcome is logged

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13. Muskan Ali — AF25LHB4561

Project: LexOps — Legal Case & Contract Compliance Management SaaS
Law firms and corporate legal departments manage active cases with deadline tracking, store and version-control contract documents, automate contract renewal and court hearing reminders, track billable hours per client and case, monitor compliance deadlines, and generate client billing statements with itemized time logs.

- **Industry Role:** Legal tech is one of the fastest-growing SaaS verticals globally. Law firms operating on paper diaries and Excel sheets are the target — every legal professional needs this
- **Free APIs:** FullCalendar.js (court hearing and deadline calendar), jsPDF (contract and billing statement generation), EmailJS free tier (hearing reminder and contract expiry notifications), Chart.js (billable hours analytics and revenue by client), Cloudinary free tier (document storage)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Partner / Associate / Paralegal / Client roles, case hierarchy, matter-level billing, contract lifecycle states
- **Real-Time Element:** Upcoming hearing countdown displayed live, billable hours total updates as attorney logs time entries, compliance deadline urgency color-coding changes in real-time as dates approach, contract expiry alerts push to dashboard without page reload

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15. Laiba Javaid — AF25LHB4563

Project: AgriSense — Precision Agriculture & Crop Market Intelligence SaaS
Farmers and agribusinesses monitor real-time weather conditions for their fields, receive crop disease outbreak alerts based on humidity and temperature patterns, track live market prices for their crops across major commodity exchanges, manage irrigation schedules, log fertilizer and pesticide application records, and get yield predictions for the season.

- **Industry Role:** Pakistan is an agricultural economy. Connecting farmers to real-time data and market prices directly reduces post-harvest losses and increases profitability — massive government and development sector interest
- **Free APIs:** Open-Meteo API (completely free — hyper-local weather, no key required), OpenWeatherMap free tier (agricultural weather alerts), FAO STAT API (free — international crop market data), Chart.js (yield trend and market price graphs), Leaflet.js + OpenStreetMap (farm location mapping), EmailJS (crop alert notifications)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Farmer / Agribusiness / Agronomist roles, field management per farm, crop calendar, market price monitoring
- **Real-Time Element:** Live weather widget for registered farm location updates every 30 minutes, commodity price ticker auto-refreshes from FAO/market APIs, disease risk alert level changes dynamically based on weather threshold crossing

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16. Areeba Kanwal — AF25LHB4564

Project: EduAnalytics — Student Performance Intelligence SaaS for Institutions
Schools, colleges, and universities track student attendance, assignment submission rates, quiz performance, and grade trends across all classes in a central analytics platform. The system flags at-risk students (those showing declining patterns) for early intervention, generates teacher performance reports, and gives institution heads a live academic health dashboard.

- **Industry Role:** Educational institutions globally are under pressure to improve student retention and outcomes. Data-driven early warning systems reduce dropout rates — directly relevant to every institution including RIPHAH itself
- **Free APIs:** Chart.js (performance trend charts, cohort analysis), jsPDF (student progress reports and parent letters), FullCalendar.js (academic calendar and exam schedule), EmailJS (at-risk student alerts to advisors), QRCode.js (student ID card generation)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase Realtime DB
- **SaaS Architecture:** Admin / Teacher / Student / Parent roles, section-level performance tracking, intervention workflow management
- **Real-Time Element:** Attendance percentage updates live as teacher marks roll call via Firebase, at-risk student count on admin dashboard increments immediately when a student crosses the threshold, grade entry triggers instant GPA recalculation visible to both teacher and student

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17. Sana Saddique — AF25LHB4565

Project: GigDesk — Freelance Workforce Operations & Compliance SaaS
Platforms, agencies, and enterprises that hire gig workers manage worker profiles and skill ratings, track hours worked per project via a built-in time tracker, calculate earnings with tax deduction summaries, manage NDAs and contracts with e-signature workflows, ensure labor compliance documentation is current, and automate payment batch processing with detailed payslips.

- **Industry Role:** The gig economy employs 1.57 billion workers globally. Every company engaging freelancers needs compliance infrastructure — legal liability for misclassification is massive
- **Free APIs:** GitHub API (developer gig worker skill verification via repo activity), Cloudinary free tier (contract document storage), jsPDF (payslip and tax summary generation), Chart.js (earnings trend and hours analytics), EmailJS (payment notification and contract alerts), Open Exchange Rates (multi-currency payout calculation)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Client company / Gig worker / Finance manager roles, project-level time tracking, compliance document vault, payment batch management
- **Real-Time Element:** Time tracker ticks live in browser as gig worker logs hours, earnings total updates in real-time as hours are submitted, contract signature status changes push live notifications to both parties

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18. Muhammad Uzair — AF25LHB4568

Project: StockDesk — Equity Research & Portfolio Risk Analytics SaaS
Individual investors and wealth managers track live stock prices, run fundamental analysis (P/E ratio comparison, EPS growth, debt-to-equity), monitor sector rotation trends, assess portfolio risk concentration, get earnings season alerts, and backtest simple investment strategies against historical data with performance metrics.

- **Industry Role:** Retail investor participation is exploding globally. Affordable equity research tools that democratize analysis previously available only to institutional investors are in massive demand
- **Free APIs:** Alpha Vantage API (free tier — real-time and historical stock data, fundamentals), Polygon.io free tier (live stock price updates), Financial Modeling Prep API (free tier — financial statements, ratios), Chart.js (candlestick-style and line charts), jsPDF (portfolio report export)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Individual investor / Wealth manager / Research analyst roles, portfolio management, watchlist system, research note workspace

- **Real-Time Element:** Live portfolio valuation updates every 60 seconds via Alpha Vantage polling, earnings alert badge fires when calendar event is within 48 hours, risk concentration gauge recalculates live as positions are added or removed

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19. Muhammad Usman — AF25LHB4569

Project: StartupRadar — Startup Ecosystem Intelligence & Deal Flow SaaS
Venture capitalists, accelerators, and angel investors track startup funding rounds globally filtered by sector, stage, and geography. The platform shows traction metrics, founder profiles linked to LinkedIn/GitHub, competitive landscape maps, deal flow pipeline management, and investment committee reporting tools.

- **Industry Role:** Pakistan's startup ecosystem raised over $400M in recent years. Local VCs and accelerators like i2i, Fatima Gobi, and Invest2Innovate need deal flow intelligence tools — currently managed via spreadsheets
- **Free APIs:** GitHub API (founder technical background verification), Crunchbase-equivalent via public data aggregation, NewsAPI free tier (startup news and funding announcements), Reddit Entrepreneur/Startup APIs (community traction signals), Chart.js (funding trend and sector distribution charts), Leaflet.js (geo-distribution map of startups), jsPDF (investment memo generation)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** VC firm / Angel investor / Startup founder roles, deal pipeline stages, investment thesis tagging, portfolio company tracking
- **Real-Time Element:** New funding round alerts push to investor dashboard via AJAX when keyword matches saved thesis, deal pipeline stage changes update the Kanban view live for all firm members, portfolio company latest news feeds refresh automatically

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20. Syed Muhammad Izhaan Ali — AF25LHB4570

Project: TrustScore — Vendor Due Diligence & Supply Chain Risk SaaS
Procurement and compliance teams assess potential vendors and partners before onboarding them. The platform aggregates public data signals — business registration status, legal case mentions, negative news, financial stress indicators, and cybersecurity exposure — into a single vendor risk score. Existing vendors are monitored continuously for score changes.

- **Industry Role:** Third-party risk management is now a regulatory requirement for banks, insurance companies, and large corporates. This is a tier-one compliance need, not optional
- **Free APIs:** Companies House UK API / SECP data scraping (free — business registration verification), AbuseIPDB API free tier (vendor website/IP reputation), HaveIBeenPwned API (free — vendor email domain breach check), NewsAPI free tier

(negative news monitoring for vendor names), CIRCL CVE API (vendor tech stack vulnerability exposure), Chart.js (risk score history trend)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Enterprise buyer accounts, vendor profiles, automated risk scoring engine, alert subscription management, audit trail logging
- **Real-Time Element:** Vendor risk score recalculates and re-renders live when any new signal data arrives, news monitoring alerts push to dashboard when vendor name appears in negative news, breach alert fires immediately when HaveIBeenPwned returns a hit for a monitored vendor domain

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21. Hannan Amin — AF25LHB4571

Project: EventIntel — Corporate Event ROI & Delegate Engagement SaaS
Conference and corporate event organizers manage delegate registrations, generate QR-based check-in passes, track session attendance in real-time via QR scanning, monitor live networking connection data, run live audience polls with instant result visualization, calculate post-event ROI based on lead generation, and produce sponsor value reports.

- **Industry Role:** Pakistan's event industry manages hundreds of corporate conferences, expos, and summits annually. Every organizer needs digital engagement infrastructure — this replaces expensive international tools like Eventbrite + Slido
- **Free APIs:** QRCode.js (free — QR pass generation and check-in scanning), FullCalendar.js (session schedule management), Chart.js (attendance heatmaps and engagement analytics), EmailJS free tier (automated registration confirmation and reminder emails), Firebase Realtime DB (live poll results), jsPDF (sponsor value and ROI report)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase
- **SaaS Architecture:** Organizer / Sponsor / Speaker / Delegate roles, event-level workspaces, session management, live engagement tools
- **Real-Time Element:** Session attendance counter increments live as delegates scan QR codes via Firebase, live poll results chart animates as audience votes come in without any refresh, check-in dashboard shows real-time entry flow with timestamps

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22. Muhammad Ahtasham — AF25LHB4572

Project: SupportFlow — Omnichannel Customer Support Intelligence SaaS
Businesses unify customer support tickets arriving from email, web form, and chat into a single inbox. The platform auto-prioritizes tickets based on SLA tier and sentiment analysis of message text, routes to the right agent team by issue category, tracks resolution times, fires SLA breach

alerts before they happen, scores customer satisfaction via automated post-resolution surveys, and gives support managers a live team performance dashboard.

- **Industry Role:** Customer support operations are the brand frontline for every business. CRM platforms like Zendesk charge enterprise pricing. This targets the Pakistani SME market with 90% of the same functionality
- **Free APIs:** EmailJS free tier (ticket intake via email and satisfaction survey dispatch), Chart.js (resolution time, volume, and satisfaction trend analytics), Firebase Realtime DB (live agent-to-ticket assignment and status updates), jsPDF (SLA compliance reports), Sentiment analysis via open-source JavaScript library (Sentiment.js — free)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase
- **SaaS Architecture:** Admin / Team Lead / Agent / Customer roles, SLA tier management, ticket routing rules, satisfaction survey engine
- **Real-Time Element:** New ticket notification appears on agent dashboard via Firebase in under 2 seconds of submission, SLA breach countdown timer ticks live on each ticket card, team performance metrics on supervisor dashboard update live as agents resolve tickets

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23. Fahad Shabir — AF25LHB4574

Project: DeFiScope — Blockchain Portfolio & DeFi Protocol Analytics SaaS
Crypto investors and DeFi participants track multi-wallet portfolio value across protocols, monitor yield farming and liquidity pool positions, receive liquidation risk alerts when collateral ratios approach thresholds, analyze gas fee patterns to optimize transaction timing, and benchmark their DeFi strategy performance against market indices.

- **Industry Role:** DeFi TVL exceeds $100 billion. Pakistani crypto investors have zero local analytical tools — entirely served by expensive international platforms. This fills a massive gap
- **Free APIs:** CoinGecko API (completely free — 10,000+ token prices, DeFi protocol data, no key required), Etherscan API (free tier — wallet transaction history and token balances), DefiLlama API (completely free — protocol TVL and yield data), Chart.js (portfolio value and yield trend charts), Open Exchange Rates (crypto-to-fiat conversion)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Investor accounts, multi-wallet address management, protocol position tracking, alert rule engine per asset
- **Real-Time Element:** Portfolio total value updates every 60 seconds from CoinGecko live prices, liquidation risk gauge animates as collateral ratio changes, gas fee recommendation index updates live showing optimal transaction windows

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24. Muhammad Abdullah Khan — AF25LHB4575

Project: TalentMarket — Developer Skills Demand Intelligence SaaS
Tech companies, HR teams, and career coaches analyze the live job market to understand which programming skills are in highest demand, what salaries different technology stacks command, how competitor companies are hiring, and where the supply-demand gaps are. Developers use it to make career investment decisions on which skills to learn next.

- **Industry Role:** Every tech company's HR strategy and every developer's career path depends on market intelligence that is currently only available to large enterprises with dedicated research teams
- **Free APIs:** GitHub Jobs / GitHub Trending API (free — real-time developer demand signals), HackerNews Jobs API (completely free — tech hiring posts), Stack Overflow Trends (public data), Reddit r/cscareerquestions API (free — community salary and skill discussions), Chart.js (demand heatmaps, salary distribution charts), FRED API (free — labor market economic indicators)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Company HR / Individual developer / Career coach roles, skill taxonomy management, market trend history database
- **Real-Time Element:** Top trending skills leaderboard updates live as new job posts are scraped, salary benchmark ranges animate as new data points are added, skill demand momentum score recalculates and reranks the leaderboard in real-time

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25. Alauddin Khan — AF25LHB4576

Project: AdPulse — Digital Campaign Performance & Budget Optimization SaaS
Marketing managers track the performance of their digital advertising campaigns across multiple channels (Google, Facebook, LinkedIn — using manually input metrics or CSV uploads), calculate true ROAS and CAC, receive budget pacing alerts when spend velocity will overshoot the monthly budget, A/B test creative performance, benchmark their metrics against industry averages, and generate agency-ready performance reports.

- **Industry Role:** Digital advertising spend in Pakistan exceeds $200M annually. Marketing managers at every company need a unified analytics dashboard that agencies currently keep opaque to retain control
- **Free APIs:** Chart.js (ROAS trend, funnel conversion, channel contribution charts), jsPDF (agency and board campaign performance reports), Open Exchange Rates (multi-currency campaign budget management), NewsAPI (industry news feed for campaign context), FullCalendar.js (campaign flight schedule management)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter
- **SaaS Architecture:** Brand / Agency / Media buyer roles, campaign hierarchy, creative asset library, budget allocation engine
- **Real-Time Element:** Budget pacing alert fires live when daily spend projection exceeds monthly target, ROAS gauge updates immediately as performance figures are entered, A/B variant winning probability score recalculates with animated chart on each data update

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26. Aown Haider — AF25LHB4577

Project: APIGuardian — Real-Time API Health & Uptime Monitoring SaaS
Development teams and CTOs monitor all their deployed APIs and web services from a central dashboard. Each endpoint is pinged at configurable intervals, response time and status codes are logged, downtime incidents are automatically detected and alerted, SLA uptime percentage is calculated per service, and historical performance reports are generated for clients or investors.

- **Industry Role:** Every tech company deploying APIs needs monitoring infrastructure. Tools like PagerDuty and Datadog cost thousands of dollars annually. This directly targets the growing Pakistani startup and software house sector
- **Free APIs:** ipify API (free — outbound IP verification), SSLLabs API (free — SSL certificate expiry monitoring), ip-api.com (free — endpoint geolocation for multi-region monitoring), Chart.js (response time trend and uptime percentage charts), EmailJS (downtime alert emails), jsPDF (SLA compliance report export for clients)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase Realtime DB
- **SaaS Architecture:** Organization accounts, service registry per team, alerting contact groups, incident management workflow, client-facing status page generation
- **Real-Time Element:** Endpoint status indicators flip from green to red in real-time the moment a ping fails via Firebase, response time graphs update live with each check cycle, incident timer starts counting the moment downtime is detected, public status page reflects service health without any manual update

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27. Muhammad Awais — AF25LHB4861

Project: ProcureIQ — Strategic Procurement & Supplier Intelligence SaaS
Procurement managers manage the full vendor lifecycle — from onboarding with document collection, through purchase order management, to delivery confirmation and performance scoring. The platform provides spend analytics by category and department, tracks contract expiry, automates three-bid comparison workflows, and gives CPOs a strategic sourcing dashboard showing cost-saving opportunities.

- **Industry Role:** Procurement is typically 60-70% of a manufacturing company's cost base. Even a 2% procurement efficiency gain on a $10M spend base saves $200K. This is one of the highest-ROI SaaS tools for any industrial company
- **Free APIs:** Chart.js (spend analytics, supplier concentration, savings trend charts), jsPDF (purchase orders and RFQ document generation), FullCalendar.js (contract expiry and delivery calendar), EmailJS (PO and RFQ dispatch emails), Cloudinary free tier (vendor document vault), Open Exchange Rates (multi-currency PO pricing)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter

- **SaaS Architecture:** CPO / Procurement officer / Supplier / Finance approver roles, PO approval workflow, contract lifecycle management, supplier performance scoring engine
- **Real-Time Element:** Live pending approval counter on CPO dashboard increments when new POs await sign-off, spend-to-budget progress bars update instantly on PO approval, supplier scorecard recalculates live as delivery and quality ratings are submitted

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28. Muhammad Ahmad — AF25LHB4531

Project: FraudGuard — Real-Time Transaction Risk & Fraud Detection SaaS
Fintech companies, e-commerce platforms, and payment processors monitor every transaction through a risk scoring engine that checks velocity rules (too many transactions in short time), geolocation anomalies (transaction location vs. registered address), device fingerprinting, and amount pattern deviation. High-risk transactions are flagged for manual review, blocked automatically if score exceeds threshold, and analysts work a live fraud queue.

- **Industry Role:** Payment fraud losses exceed $32 billion globally. Every fintech, neo-bank, and e-commerce platform needs fraud detection infrastructure — previously accessible only to large banks at massive cost
- **Free APIs:** ip-api.com (free — transaction IP geolocation for anomaly detection), AbuseIPDB free tier (transaction IP reputation scoring), Chart.js (fraud rate trends, risk score distribution, detection accuracy charts), Firebase Realtime DB (live fraud queue updates for analyst team), jsPDF (regulatory fraud incident reports)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase
- **SaaS Architecture:** Platform admin / Fraud analyst / Risk manager roles, configurable rule engine, transaction risk scoring API module, manual review workflow
- **Real-Time Element:** High-risk transaction alert appears on analyst dashboard via Firebase within seconds of detection, live fraud rate percentage updates on the monitoring screen, analyst claim/reject actions on fraud queue reflect instantly for all team members via Firebase

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29. Amara Subhan — AF25LHB4800

Project: WorkforceWell — Employee Burnout Detection & People Analytics SaaS
HR departments and people managers track organizational health through anonymous weekly pulse surveys, monitor work hour patterns for burnout risk indicators, run sentiment analysis on open-ended feedback, measure eNPS (employee net promoter score) over time, identify flight-risk employees before they resign, and build data-driven retention strategies with intervention tracking.

- **Industry Role:** Employee turnover costs companies 150-200% of annual salary per lost employee. People analytics is the fastest-growing HR tech segment. This addresses the talent retention crisis facing every growing company

- **Free APIs:** Chart.js (burnout heatmaps, eNPS trend, survey response distribution), Sentiment.js open-source library (free — feedback text sentiment scoring), jsPDF (HR board people analytics report), EmailJS free tier (pulse survey dispatch and response reminders), Firebase Realtime DB (live survey response collection)
- **Tech Stack:** HTML5, Bootstrap 5, JavaScript, PHP, MySQL, CodeIgniter, Firebase
- **SaaS Architecture:** CHRO / HR Business Partner / Manager / Anonymous employee roles, department-level health tracking, intervention case management, anonymity-preserving data aggregation (minimum N=5 before revealing)
- **Real-Time Element:** Live survey response rate counter updates as employees submit via Firebase, burnout risk score for each department recalculates immediately when new responses arrive, eNPS gauge animates on the HR dashboard as scores are logged without page reload

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Summary Table — Advanced Industry Projects

Sr #	Student	Project	Industry Vertical	Key Free APIs
1	Mariam Sana Tariq	TalentPulse	HR Tech	GitHub, Hunter.io, Firebase
2	Eman Fatima	BrandSentinel	MarTech / PR	Reddit, HackerNews, NewsAPI
3	M. Saib Bashin	LogiTrack	Supply Chain	Leaflet, OSRM, Nominatim
4	Muhammad Bilal	PriceIntel	E-Commerce	Alpha Vantage, Chart.js
5	Ali Azam	CarbonLedger	ESG / CleanTech	CO2 Signal, World Bank
6	Mubeen Ali Zulfiqar	CyberPulse	Cybersecurity	CIRCL CVE, AbuseIPDB, Shodan
7	Abdul Subhan	FleetOps	Logistics / Last-Mile	Leaflet, OSRM, Firebase
8	Ahmad Aamir	FinFlow	FinTech / SME Finance	ExchangeRate, jsPDF, EmailJS
9	Khanzada Sarim	PropIntel	Real Estate Tech	Leaflet, World Bank, jsPDF
10	Zarmeen Nadeem	HealthOps	HealthTech	FullCalendar, Firebase, QRCode.js
11	Abeeha	InfluenceIQ	Influencer / MarTech	YouTube API, Last.fm, Cloudinary
12	Amna Shoukat	GovTender	GovTech / Procurement	PPRA RSS, World Bank, EmailJS
13	Muskan Ali	LexOps	LegalTech	FullCalendar, jsPDF, Cloudinary
15	Laiba Javaid	AgriSense	AgriTech	Open-Meteo, FAO STAT, Leaflet
16	Areeba Kanwal	EduAnalytics	EdTech	Chart.js, Firebase, jsPDF
17	Sana Saddique	GigDesk	FutureOfWork / HRTech	GitHub, Cloudinary, jsPDF
18	Muhammad Uzair	StockDesk	WealthTech	Alpha Vantage, Polygon.io, FMP
19	Muhammad Usman	StartupRadar	VC / Startup Ecosystem	GitHub, NewsAPI, Leaflet
20	Syed M. Izhaan Ali	TrustScore	Risk & Compliance	AbuseIPDB, HaveIBeenPwned, NewsAPI
21	Hannan Amin	EventIntel	EventTech	Firebase, QRCode.js, FullCalendar

Sr #	Student	Project	Industry Vertical	Key Free APIs
22	Muhammad Ahtasham	SupportFlow	CRM / Customer Ops	Firebase, Sentiment.js, EmailJS
23	Fahad Shabir	DeFiScope	DeFi / CryptoTech	CoinGecko, Etherscan, DefiLlama
24	Muhammad Abdullah Khan	TalentMarket	HR Intelligence	GitHub Trending, HN Jobs, FRED
25	Alauddin Khan	AdPulse	AdTech / MarTech	Chart.js, jsPDF, FullCalendar
26	Aown Haider	APIGuardian	DevOps / InfraTech	SSLLabs, ip-api, Firebase
27	Muhammad Awais	ProcureIQ	Procurement / Supply Chain	Chart.js, jsPDF, Cloudinary
28	Muhammad Ahmad	FraudGuard	FinTech / Risk	ip-api, AbuseIPDB, Firebase
29	Amara Subhan	WorkforceWell	PeopleOps / HRTech	Sentiment.js, Firebase, jsPDF